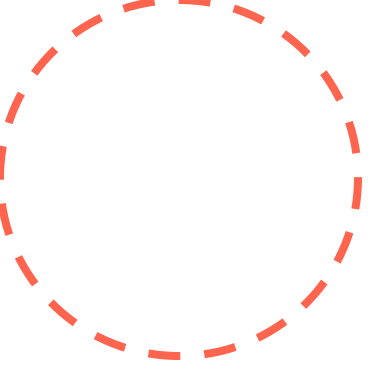
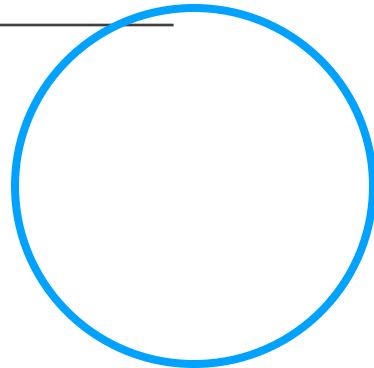


BAXI 2213 KNOWLEDGE *Based* / SYSTEM

INTRODUCTION



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Our Class

1. Timetable

2. WhatsApp Group (Any Updates)

3. Ulearn (TP, Notes, Tutorial, Quizzes, Assignments etc)

4. Attendance (Online Participation)

5. Lab Work : CLIPS / Python / Prolog

6. Assessment

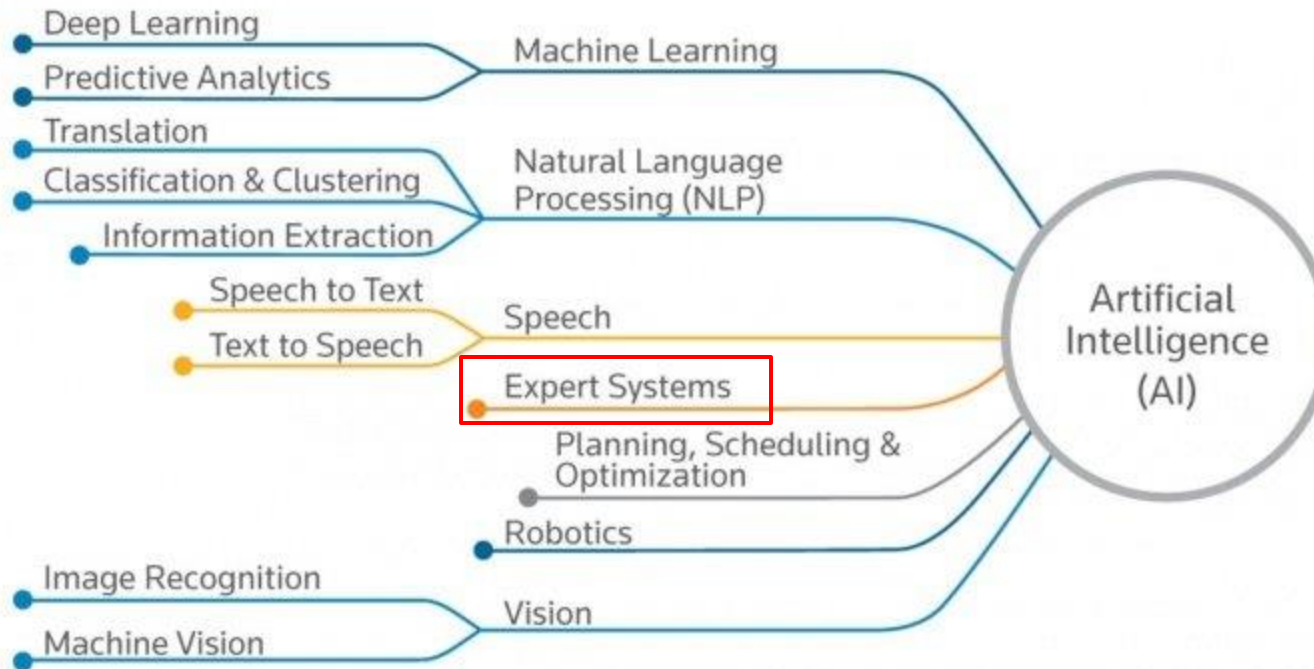
- i. Lab Test
- ii. Quiz 1 and 2
- iii. Project
- iv. Assignment
- v. Tutorials / Lab Work
- vi. Final Test / Mid Test



OUTLINE

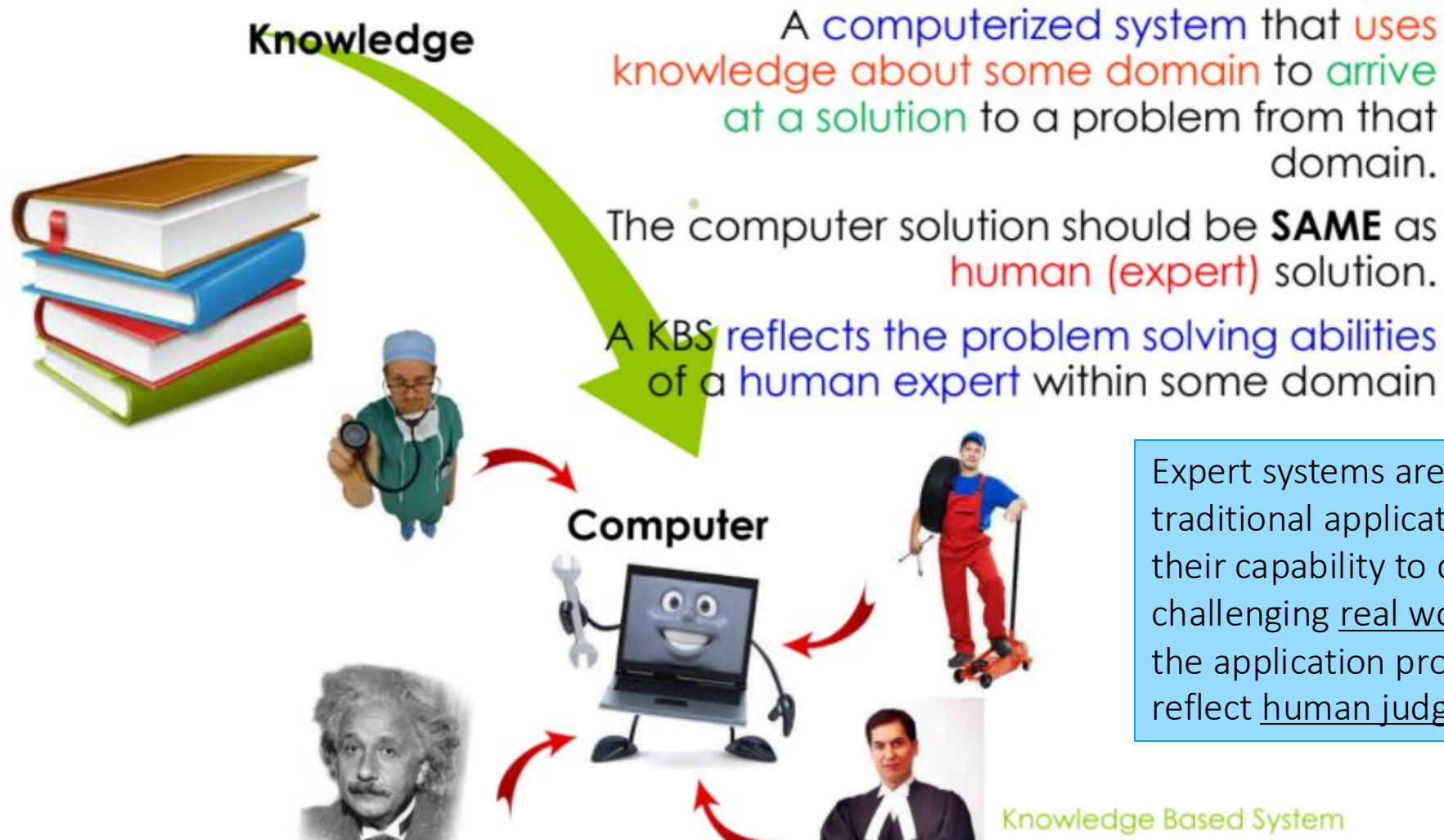
- Why you need to learn this subject?
- What is KBS?
- KBS: Early works
- DKIW
- Types of Knowledge
- Types of Experts
- KBS Structure
- KBS vs Conventional System
- KBS Lifecycle
- KBS Language, Shells & Tools
- KBS Advantage/Limitation

Why We Learn This Subject?



Historically **Expert Systems** grew with AI, and can be regarded as a branch of AI. **Artificial Intelligence (AI)** is the part of computer science concerned with designing intelligent computer systems, that is, systems that exhibit the characteristics we associate with intelligence in human behavior – understanding language, learning reasoning, solving problems and so on.

What is Knowledge Based System?



Expert systems are different from traditional application programs in that their capability to deal with challenging real world problems through the application process that reflect human judgment and intuition.

What is Knowledge Based System?

- The term *expert system* is often used synonymously.
- A computer system that emulates the decision-making ability of a human expert in a restricted domain.
- Edward Feigenbaum (Father of Expert System) said, expert system meaning is:
“An intelligent computer program that uses knowledge and inference procedures to solve problems that are difficult enough to require significant human expertise for their solutions.”



What is Knowledge Based System?



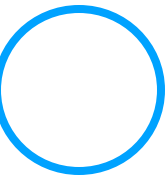
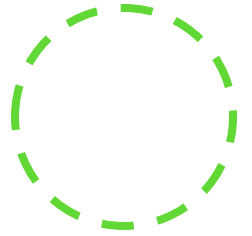
KNOWLEDGE

HOW
KNOWLEDGE
IS USED

Three fundamental concepts of KBS distinguished them from **conventional algorithmic** programs and **general search-based** programs.

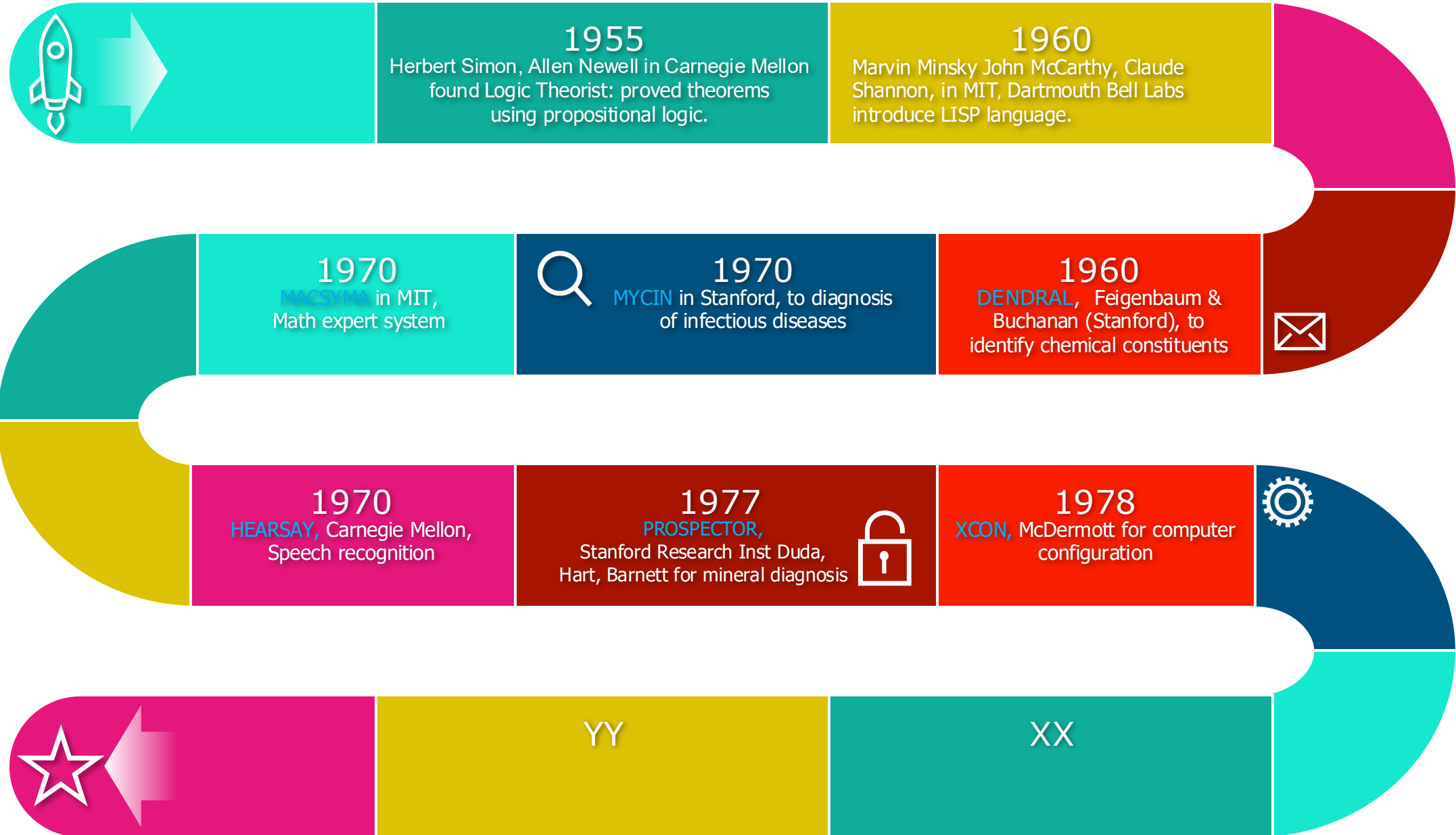


SPECIFIC[®]



What is Knowledge Based System?

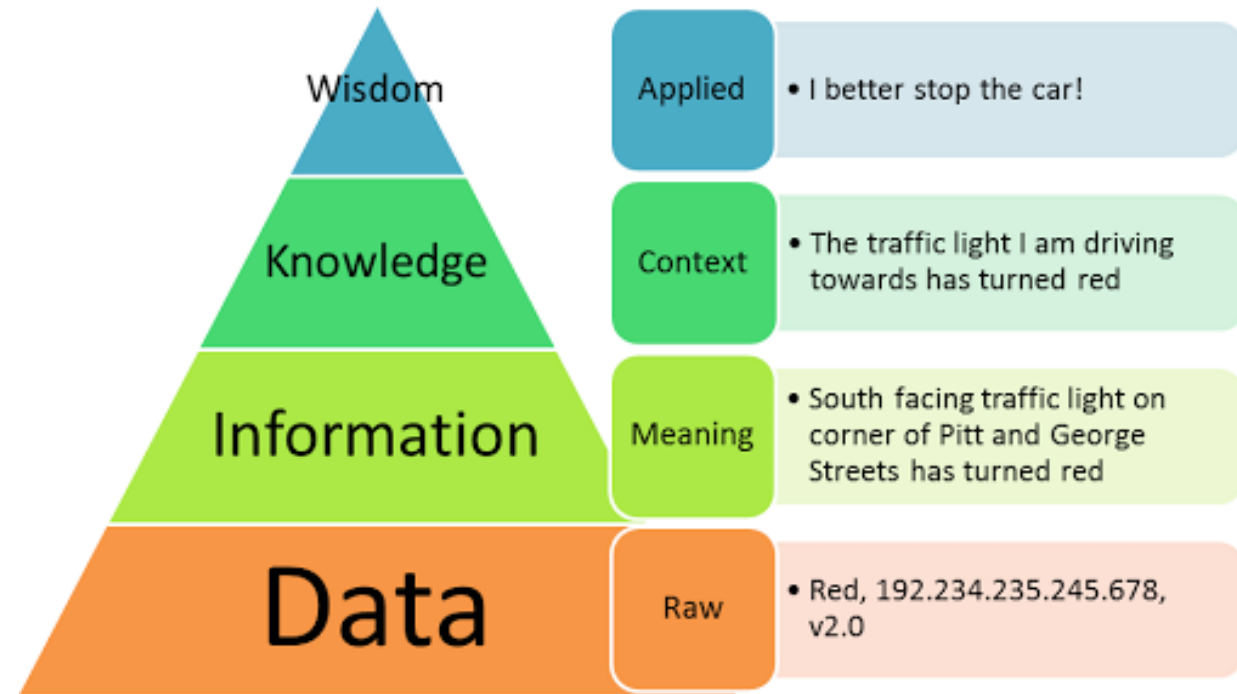




KBS Early Works

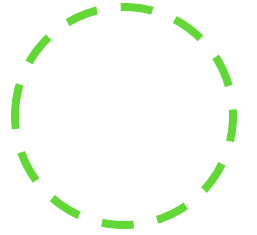
DIKW

- Data, information, knowledge and wisdom are **major elements of human thinking and reasoning process.**
- A KBS is a computer based system, which **uses and generates** knowledge from data, information and knowledge.

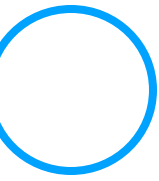




DIKW



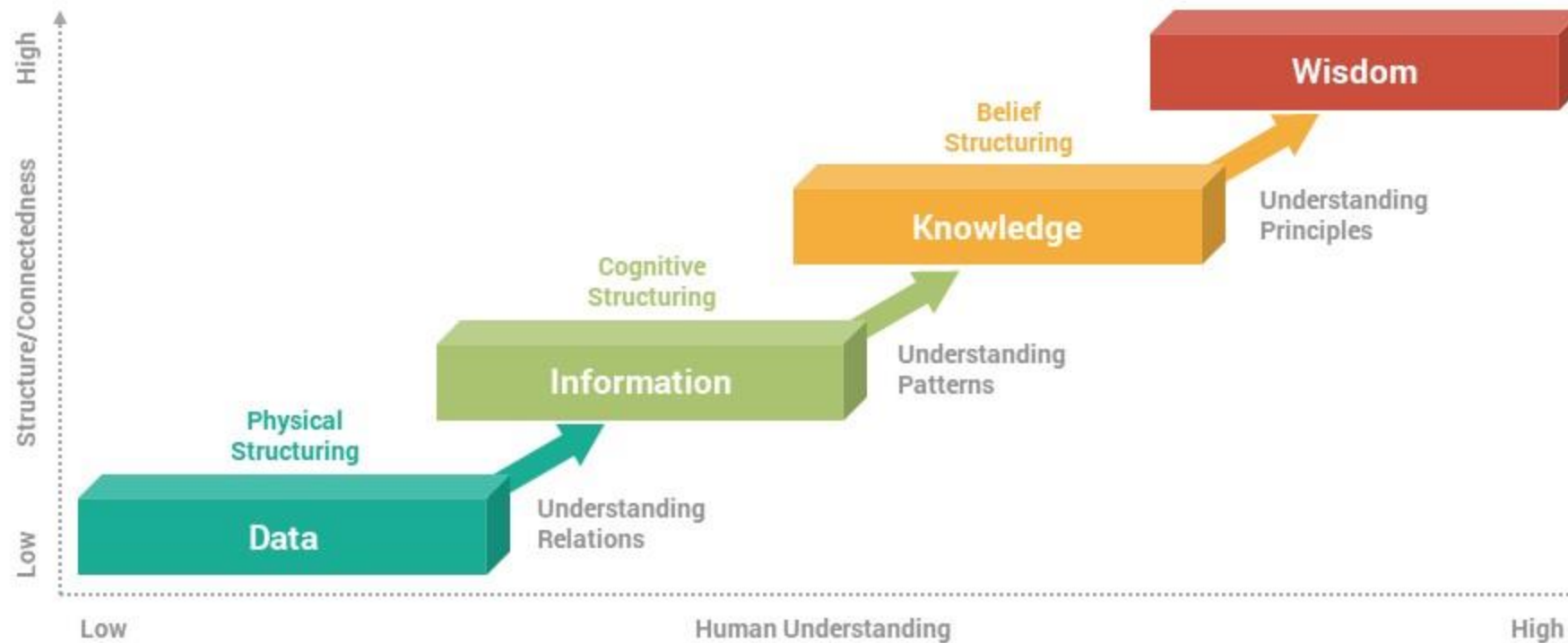
TOM CHANLEY



DIKW

DIKW Hierarchy Model Flow

DIKW Model



DIKW

DIKW Model

DIKW Model



Data

The first step is Data. Data is the starting point for reaching a meaningful result in the end. Logging, records, measurements, etc. are all data.

Information

Data is processed in the Information stage of the hierarchy. Relationships in the data are revealed, and analysis is done to find an answer to Who, What, When and Where questions.

Knowledge

The third level in the DIKW hierarchy is Knowledge. The Knowledge step aims to answer the How question. Specific measures are identified, and the information gained in the previous step is used to answer questions based on these measures.

Wisdom

The fourth and the last step in the DIKW hierarchy is Wisdom. At the Wisdom stage, the knowledge found in the previous step is applied and implemented. Wisdom is the top level reached in DIKW hierarchy and answers the Why question.

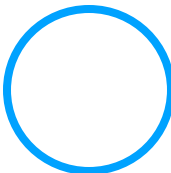
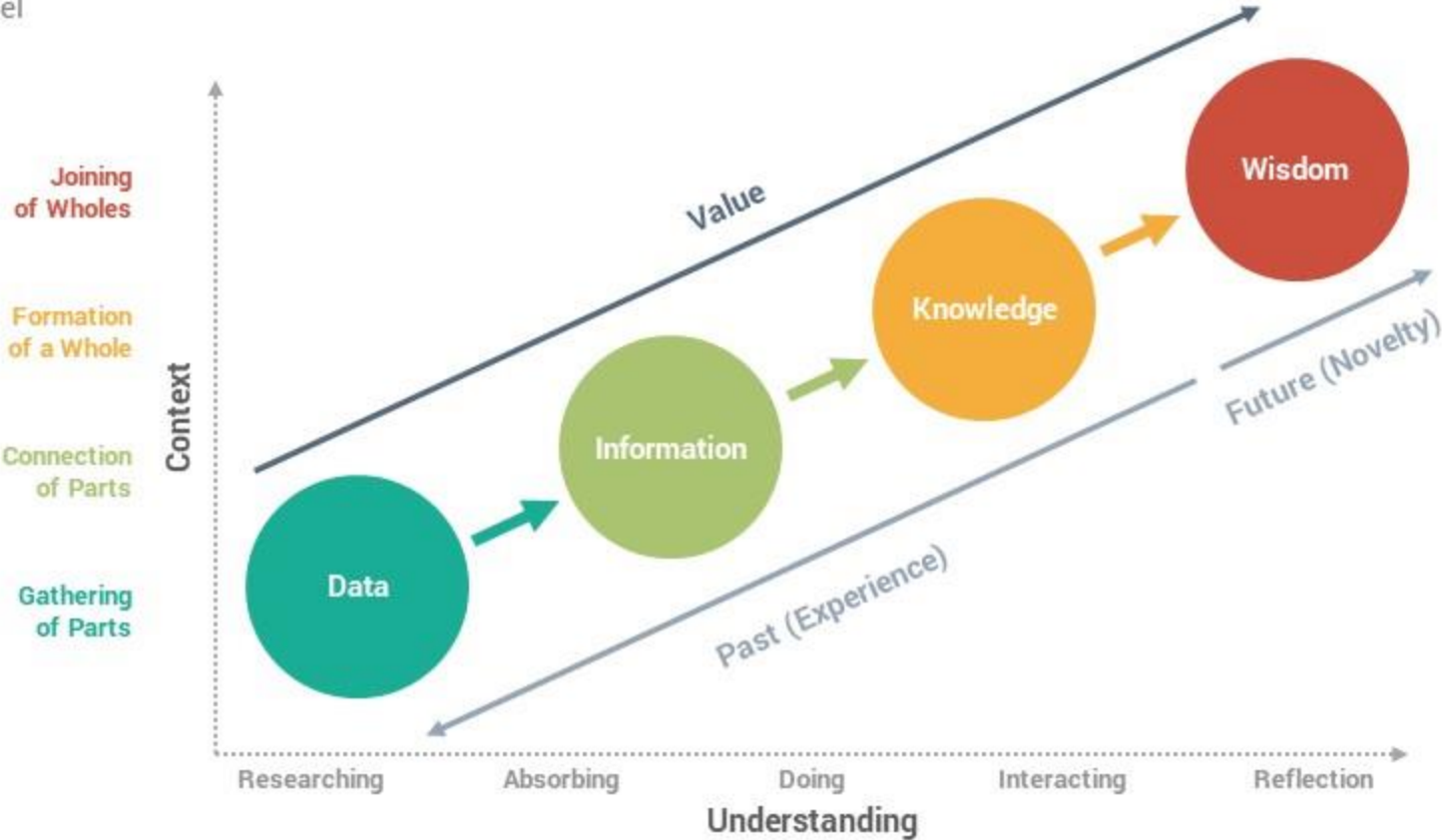


DIKW



Conceptual Relations in DIKW

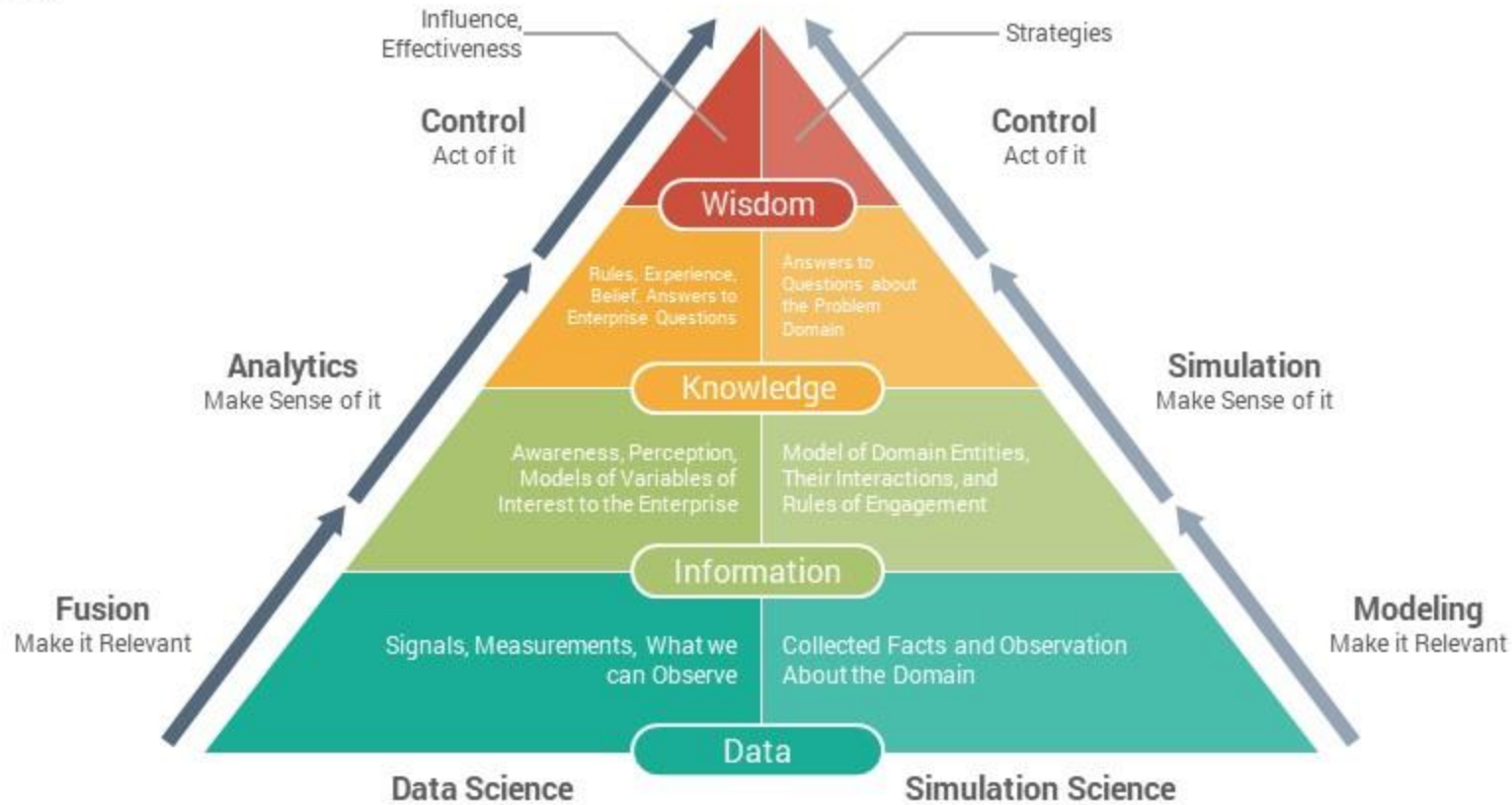
DIKW Model



DIKW

DIKW Model

DIKW Model

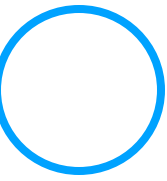
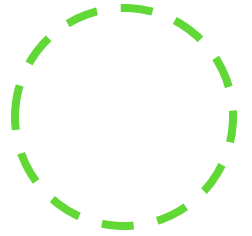
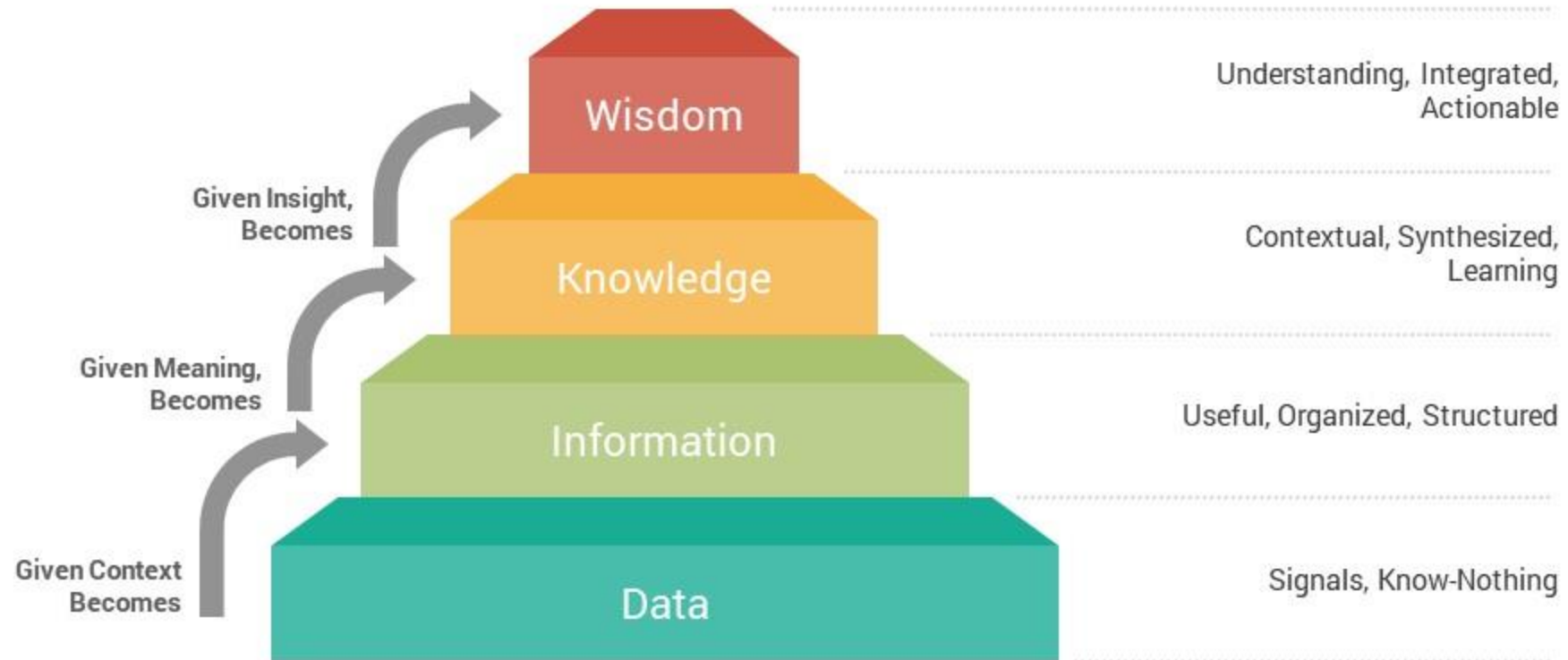




DIKW

DIKW Model

DIKW Model



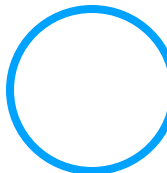
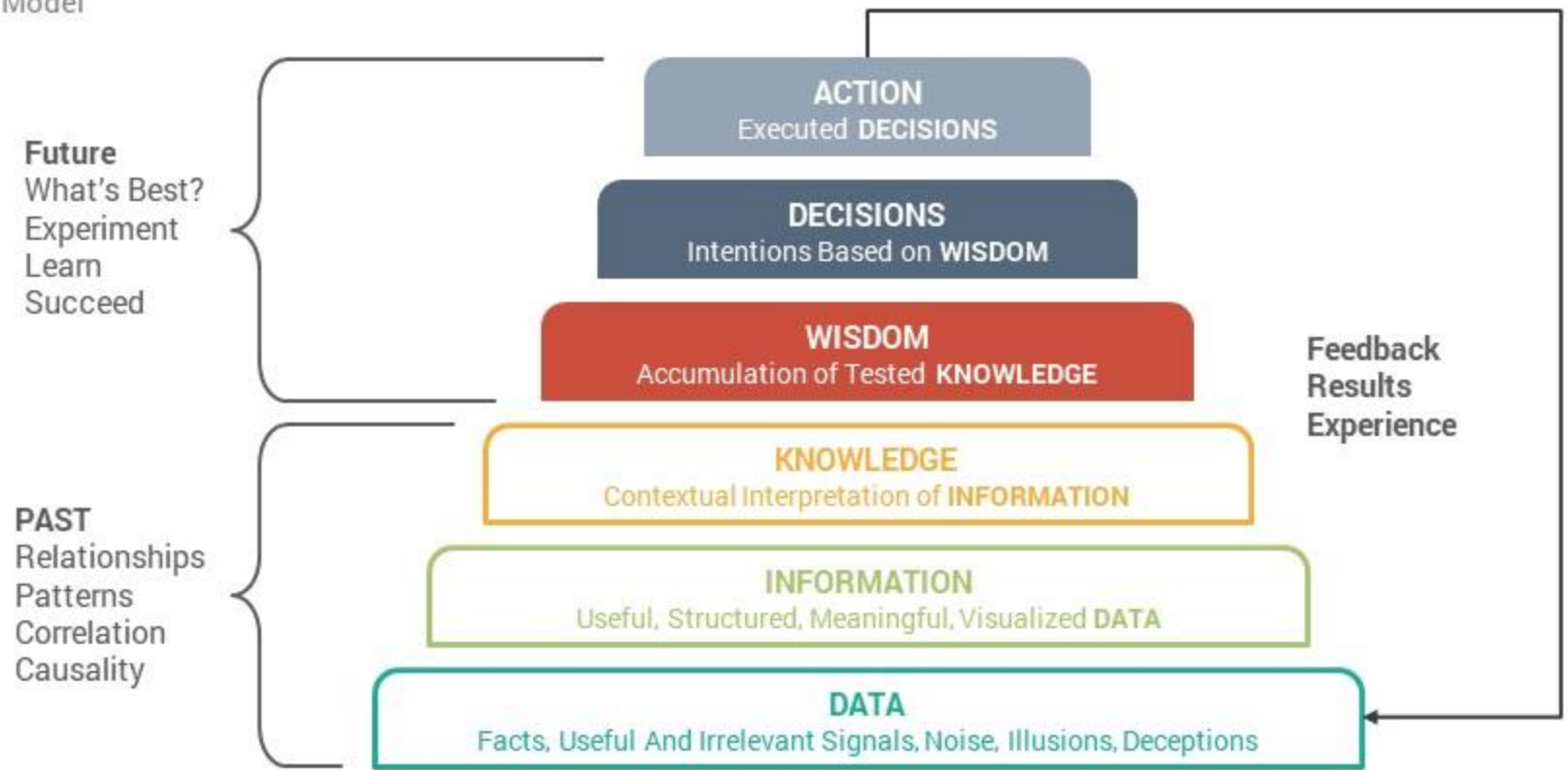


DIKW



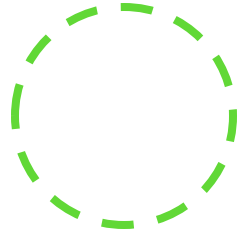
DIKW Model

DIKW Model



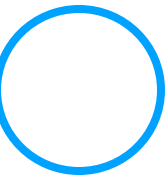
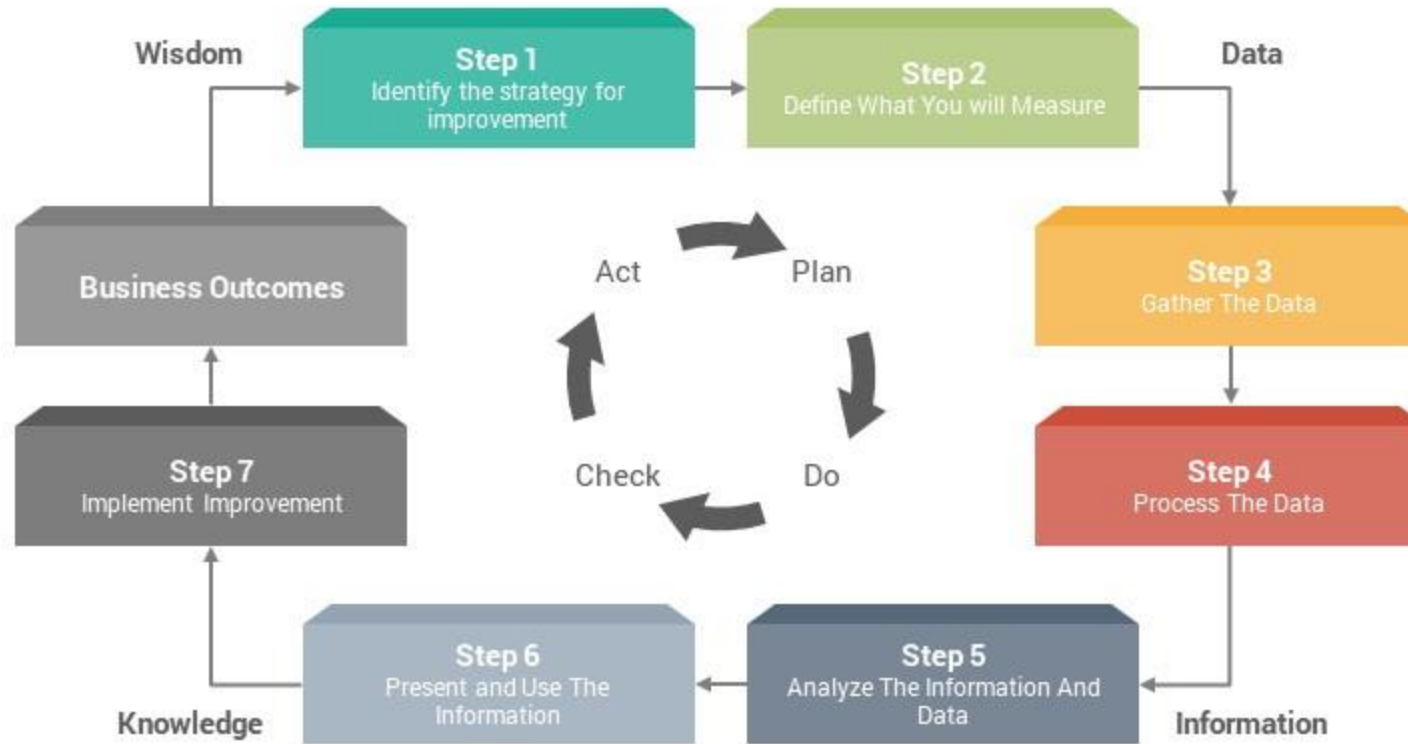


DIKW



The 7 Steps Of DIKW Model

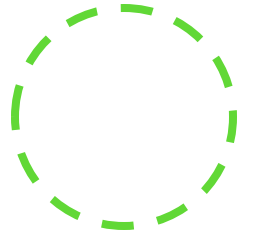
The 7 Steps Of DIKW Model





DIKW

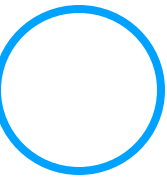
WHAT IS THIS?



1. A box (data)
2. The box is 3' wide, 3' deep, and 6' high (info)
3. The box is very heavy, and if you lift this box you will get slip disk. (wisdom)
4. When I open the box it has food in it (knowledge)
5. It is colder inside the box than it is outside (knowledge)
6. You usually find the box in the kitchen (info)
7. When you open the door the light comes on (knowledge)
8. When you move this box you usually find lots of dirt underneath it (wisdom)



This example of DIKW. The answer is in bracket. Is the answer correct?



To Be Continued.....